The complete expansion for every dimension: $H_m(L) = \sum_{i \le m} c_{m,i} \log^i L + O(1/L)$ and $Z_d(n) = n^{-1/2} P_d(\log n) + O(n^{-1})$

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Abstract

Units 41 and 42 identified the constant terms below $\log n$ and $(\log n)^2$ by hand. This unit proves the underlying transfer step once, for the iterated logarithmic primitives $H_m =$ `iterLogPrim` m of unit 39: with explicit recursively defined coefficients $c_{0,0} = A$, $c_{m+1,i+1} = c_{m,i}/(i+1)$, $c_{m+1,0} = \int_0^1 H_m/x + \int_1^\infty \theta_m/x$,

$$H_m(L) = \sum_{i=0}^{m} c_{m,i} \log^i L + \theta_m(L), \qquad |\theta_m(L)| \le \frac{M}{L} \quad (L \ge 1), \qquad \theta_{m+1}(L) = -\int_L^\infty \frac{\theta_m(x)}{x}\, dx,$$

and $c_{m,m} = A/m!$. Hence every all-equal chart integral has a complete expansion $Z_{k+2}(n) = n^{-1/2} \sum_{i \le k+1} c_{k+1,i} (\frac{1}{2} \log n + (k+2) \log b)^i + O(n^{-1})$. Zero `sorry`/`axiom`.

1 Setting

The recursion $H_{m+1}(L) = \int_0^L H_m(x)/x\, dx$ preserves the shape "polynomial in $\log x$ plus a remainder $O(1/x)$": over $[1, L]$ the polynomial integrates termwise via $\int_1^L \log^i x/x = \log^{i+1} L/(i+1)$, the remainder over $[1, \infty)$ converges to a constant that becomes the new $c_{m+1,0}$, and the tail over $[L, \infty)$ is the new remainder. The remainder bound M/L is preserved exactly because $\int_L^\infty M x^{-2} = M/L$.

2 The things formalised

```
noncomputable def iterLogCoeff ( a : ) :  →  →
  | 0, 0 => quadMass  a
  | 0, _ + 1 => 0
  | m + 1, 0 => ( x in Ioc 0 1, iterLogPrim  a m x / x)
      +  x in Ioi 1, (iterLogPrim  a m x
          -  i  Finset.range (m + 1), iterLogCoeff  a m i * Real.log x ^ i) / x
  | m + 1, i + 1 => iterLogCoeff  a m i / ((i : ) + 1)
theorem iterLogCoeff_top ... : iterLogCoeff  a m m = quadMass  a / (m.factorial : )
theorem iterLogRem_succ_eq ... (hL : 1  L)
    (hb :  L, 1  L → |iterLogRem  a m L|  M / L) :
    iterLogRem  a (m + 1) L = -  x in Ioi L, iterLogRem  a m x / x
theorem iterLogPrim_expansion ( a : ) (h : 0 < ) (m : ) (L : ) (hL : 1  L) :
    |iterLogPrim  a m L
        -  i  Finset.range (m + 1), iterLogCoeff  a m i * Real.log L ^ i|
```

```
        quadMoment   a / L
theorem iterChart_expansion_isBigO (  a b :  ) (h  : 0 <  ) (hb : 0 < b) (k :  ) :
    (fun n => iterChart   a b (k + 2) (Real.sqrt n)
        - n ^ (-(1/2 :  )) * iterChartCoeff   a b k n) =O[atTop] fun n => n ¹
```

3 Proof strategy

The coefficients are defined by structural recursion on m with the index i as a varying second argument, so all three equation lemmas are `rfl`. The remainder bound is proved by induction on m; the inductive step is the exact identity $\theta_{m+1}(L) = -\int_L^\infty \theta_m/x$, which takes the current bound as a hypothesis (it is needed for integrability of θ_m/x on $[1,\infty)$) and is proved by the same three-way split as at $d = 3$: $(0,1]$, $(1,L]$ with the pointwise decomposition $H_m/x = \sum c_{m,i} \log^i x/x + \theta_m/x$, and $\int_1^\infty = \int_1^L + \int_L^\infty$. The re-indexing $\sum_{j\le m+1} c_{m+1,j} \log^j L = c_{m+1,0} + \sum_{i\le m} c_{m,i} \log^{i+1} L/(i+1)$ is `Finset.sum_range_succ'` plus the coefficient equation. The base case is $|F(L) - A| = A - F(L) \le M/L$ from unit 35. The chart statement is unit 39's reduction $Z_{k+2} = n^{-1/2} H_{k+1}(\sqrt{n} b^{k+2})$ with $\log(\sqrt{n} b^{k+2}) = \frac{1}{2} \log n + (k+2) \log b$.

4 Roadblocks and resolutions

- The file compiled without errors on the first pass; the only fixes were deprecation renames (`continuousOn_finsetSum`, `integrable_finsetSum`, `integral_finsetSum`), an unused hypothesis, and three over-long lines.

5 What was Mathlib, what was new

Mathlib: `integral_finsetSum`, `integrable_finsetSum`, `continuousOn_finsetSum`, `Finset.sum_range_succ'`, `norm_integral_le_of_norm_le`, and the interval-to-set integrability bridge `intervalIntegrable_iff_integrableOn_Ioc_of_le`. New: the recursive coefficient scheme, the exact remainder recursion, and the uniform M/L bound for every m.

6 Lessons

Two hand-worked instances (units 41, 42) were enough to see the induction; formalising the third level as a general step cost less than either instance because the statement had already stabilised. The remainder bound being *exactly* preserved ($M/L \mapsto M/L$) is what makes the induction hypothesis clean; a bound that degraded with m would have needed the `iterConst` bookkeeping of unit 39.

7 Outcome

For every dimension $d \ge 2$ the all-equal chart integral has a complete polynomial-in-log n expansion with explicit recursive constants and $O(n^{-1})$ remainder, subsuming units 41–42. Next candidates: the weighted primitives F_γ for general (k,h), or the mixed-multiplicity case in general dimension (several minimal coordinates plus several non-minimal ones).